Data Deposit Guide for Iowa Research Online (IRO)

A. Prepare the data

1. **When possible, convert data to open, non-proprietary formats, which facilitate long-term access.**
   For instance, Excel files should be converted to comma separated values (.csv) or tab separated values (.tsv), in a text encoding standard such as UTF-8 (to facilitate cross-platform compatibility).
   - File formats info: https://www.lib.uiowa.edu/data/manage/file-formats/
   - Questions? Contact us: lib-data@uiowa.edu or brian-westra@uiowa.edu
2. **For tabular data, review the data structure and revise as needed before conversion to a .csv format.**
   These steps make it easier to convert from Excel to comma separated values (.csv) or tab-separated values (.tsv), and enable others to more easily open, view and understand your data, and for software to read/use the data.

   For instance:
   - don't use colored text or backgrounds (they will be lost in conversion to .csv files)
   - use a single column header row
   - avoid spaces, special characters, and punctuation in headings/variable names
   - in general, use columns for variables, rows for observations
   - exercise caution if you have date fields in an Excel file, and check them after conversion to .csv;
   - do not embed formulas in cells (they will not convert to .csv)
   - use a separate Excel tab for each table – don't combine tables into one large table or tab

   See these short guides for practical steps you can take:
   - Data Organization in Spreadsheets (how-to article): https://www.tandfonline.com/doi/full/10.1080/00031305.2017.1375989
   - Spreadsheet data structure (UI Libraries Website)
3. **Create a data dictionary(ies) for tabular data -** see **F.2** below
4. **Provide access to data for peer reviewers:** for data deposited in Iowa Research Online (IRO) we can provide a temporary private link to the dataset for peer review access. The link generates a .zip file consisting of all the files that you want reviewers to be able to access. Access can be removed at any time, but is usually set up for the time window that you determine (e.g., 30 days).

B. Obtain a reserved DOI from us so you can include a citation for the data in your manuscript.

Digital Object Identifiers (DOIs) are unique persistent identifiers for digital assets such as articles and datasets. They allow computer systems to link materials together, and to track your impact, such as through citations.

We can reserve a DOI for you before your data is published, so that you can include a citation and reference for the data set(s) in the manuscript, even before the data is made openly accessible*.

- See also sections **C** (cite your data in the manuscript) and **D** (data availability statement) below.

*The DOI will not work until it is activated. Once it is activated, the dataset is published, and this is not reversible.

To reserve a DOI for your data, we need just these two things:

- The Title of the dataset (**B.1**) and
- The name of at least one UI faculty member **or** research assistant/staff who is a dataset creator (part of **B.2** below)

B.1. Title of the Data Set:

***Not* the same as the title of the article.**

Create a title that is succinct and conveys that this is a dataset.

A title that simply adds "Dataset:" or "Data for…" to the title of the related article is okay but it's even better if it conveys more information about the dataset as a unique output of your research.

Examples:

- ***PCB Emissions data from Paint Colorants***
- **Hepatic expression of Cytochrome P450 enzymes in mice exposed to 3,3'-dichlorobiphenyl (PCB 11).**
- **A spatial and projection-based transcriptomic atlas of paraventricular hypothalamic cell types**

* The dataset title is also included in the README.txt file (see **F.1**)

Title for the data/dataset:

The rest of these details (**B.2** through **B.6**) are ***required*** *before your data can be published*

* These details are usually included in the README (**F.1** below) and we can copy them from the readme if you provide them there.

B.2. List all Creators ("Authors") of the dataset, in the order you want them to be listed when citing the data.

* this is usually included in the README.txt file, and we can copy it from there (see **F.1**).

Please provide first name (and middle initial if so desired), and last name.

For UI faculty and staff, if you have an ORCID and it is connected to the UI, all we need is your name.

- If you don't have an ORCID yet, start here: https://iam.uiowa.edu/planter/
- If you have an ORCID, connect it to the University of Iowa: https://guides.lib.uiowa.edu/ORCID

For graduate students: please include ORCIDs

For creators/authors from other institutions, please provide their ORCIDs if you can.

Use this table to list the Creators, in the order that you want them to appear in the citation for the dataset:

Creator Name	**ORCID** (if known, for UI students and non-UI co-creators.	**Institution** (for non-UI researchers)

B.3. Publication Date

We will enter the year, month and day when the dataset is published in IRO

B.4. Funder(s) and funder-supplied grant number(s)

* this is usually included in the README.txt file and we can copy it from there.

Funder	**Grant number (if NIH, see the guidelines for NIH grant numbers)**

B.5. Indicate which license you would like to use for the data.

* this is usually included in the README.txt file and we can copy it from there.

We recommend Creative Commons (CC) licenses for data sharing. Some common options are: CC BY V4.0, and CC 0.

- Once a CC license is chosen and the dataset is published, the license cannot be changed, so please review the licenses on the CC website before choosing.

CC-BY V4.0 among other things, requires others to give you credit if they use your data.

CC 0 puts no restrictions on what others can do with your data; it puts your work into the worldwide public domain.

Other CC licenses include one or more other requirements for users of your data. These are described on the Creative Commons website

- If you're not sure, contact us: lib-data@uiowa.edu or brian-westra@uiowa.edu, or use the CC License Chooser

Your license choice for data:

B.6. License(s) for other materials you might share.

If you want to share other kinds of materials in the same deposit as the data, we can assign different licenses for each within the same deposit.

- If you want to share software or code/scripts, there are other options too, so please contact us: lib-data@uiowa.edu or brian-westra@uiowa.edu

For software/code licenses:

Want to keep it simple? Start here:

- Choose an open-source license - Simplified license chooser for some common licenses.

More extensive lists of licenses for software:

- Open Source Initiative (OSI) Approved Licenses - Search and filter a large list of OSI-approved options
- System Package Data Exchange (SPDX) - Includes OSI approved licenses and others

Your license choice for code/software:

For copyrightable works such as instructional videos, select a Creative Commons license

- CC license chooser: https://creativecommons.org/choose/

Your license choice for instructional materials, videos, etc.:

C. Cite the data in the text of your article manuscript/thesis and include it in the references.

Once you have the reserved DOI from us (**B** above), you can create the citation.

- This will ensure that the citation is displayed in all formats of your article, and enables systems to track citations of your data.

The precise order and punctuation will depend on the citation style of the journal in which you are publishing. We recommend including the identifier if possible since that is the most unique element of the data citation, and enables others to navigate to the record.

These are the most common elements of a data citation:

Creator (PublicationYear). Title. Publisher. [resourcetype]. Identifier

- **Creator** (the "authors" of the dataset)

- **Publisher** is the University of Iowa, since it is being published in the UI's repository.
- **resourcetype** is usually 'dataset' but might also be 'collection,' 'model,' 'software,' etc., depending on the nature of the material being deposited
- **Identifier** is https://doi.org/ + DOI

* The resourcetype enables computer systems to identify this as a dataset citation. Not all citation styles incorporate this into their guides, but include it if you can, in brackets or parentheses, depending on citation style.

Example:

Jahnke, Jacob C. and Hornbuckle, Keri C. (2019): Dataset for PCB Emissions from Paint Colorants. University of Iowa. [Dataset]. https://doi.org/10.25820/vtd8-n771

D. Include a Data Availability Statement in your article manuscript

Note: This is in addition to citing the dataset as described in **C** above.

In the text of the document, include a statement describing how the data underlying the findings of your article can be accessed. This should include a reference to the citation for the dataset, in the article's references section. **

- Your journal or publisher might provide guidelines for the data availability statement. For example, here are Taylor & Francis examples of data availability statements.

** Some publishers may also have an author submission process that includes a form or fields for a data availability statement. This should be filled out as well, but often this information is only displayed with the online version of the article, so it is not sufficient.

E. Do you need to share the data with peer reviewers before it is published?

Contact us and we can create a private URL that you can share with peer reviewers. The link will download a zip file of the files that you want reviewers to access. The link will work for a time period you specify (e.g., 30 days), and can be deactivated at any time.

F. Before publication of the dataset, please provide the following details about the dataset:

These are required or highly recommended to enable discovery, reuse, and citation of your data.

F.1. Create a README.txt file. REQUIRED

The README file provides context about the data, explains the methods, and is indexed by search engines for use in web searches. **The more detail, the better**.

Cornell University's Research Data Management Service Group offers a great outline.

For more information, see: https://www.lib.uiowa.edu/data/share/metadata/#readme

a. **Save the txt file with UTF-8 encoding**. This helps avoid problems with opening and reading the file.

b. In the list of files, list the README file first, then the data files. Readers will usually want to look at the README first.

c. If you create data dictionary files (**2, below**) list them right after the README, or if there are multiple README files, group then with the data file(s) they describe.

d. If you have more than 10 files, it can be helpful to **incorporate a number prefix into the file names**, such as:

   – 01_README.txt
   – 02_datadictionary.csv
   – 03_Substance_Descriptors_and_Identifiers.csv
   – 04_toxdata.csv
   – 05_behavioraldata.csv
   – etc

e. **If you have a large number of files**, describe the file naming convention (if you have one), and/or describe how the files are named, organized and relationships between them. Grouping similar files together into .zip archives is often helpful.

f. **For Iowa Superfund Research Program data deposits**: list and describe the chemical substances file (**see F.6 below**).

F.2. For tabular data, create a Data Dictionary as a separate file (.csv, or a .txt file) for each data file or set of data files. HIGHLY RECOMMENDED.

A data dictionary explains what the variable names are and provides other details about them for other researchers.

Data dictionaries usually include these kinds of information:

- Variable name/column heading
- Description of the variable name/column heading; may include a definition/description of the variable)
- Measurement units where applicable (e.g., grams, hours, milliliters, pg/m3)
- Allowed values, or range of values, if applicable
- Are null values allowed for the variable? How are null values expressed?
- Other codes and abbreviations (e.g., for missing data, data below limit of detection/quantitation)
- The data type of each entry (text, string, number, ISO 8601 date, etc.)
- Synonyms for the variable name
- Other resources

For more information, see https://www.lib.uiowa.edu/data/share/metadata/#dictionaries and also: Open Science Framework, the Smithsonian, and this USDA blank template.

F.3. For surveys, create a Codebook, and share the survey instrument. HIGHLY RECOMMENDED

Codebooks are a type of data dictionary that are more appropriate for survey and interview data.
See: https://www.lib.uiowa.edu/data/manage/documenting/README/#codebooks

F.4. Create an abstract and a methods statement (if you did not include these in the README.txt file). REQUIRED

The abstract should be a brief description of the data and the context in which the data was collected or created.

Focus on the data, rather than reusing the abstract from the related article.

Abstracts are indexed by search engines, so they help make your data more discoverable, and they provide context and information about the dataset for the researcher who finds your data.

An abstract is usually included in the README.txt file (above). If it is in your README file, we will use that for the Abstract when we create the data record.

Abstract, if it is not in the README file:

Methods can also provide important information to researchers and others who may find and view your data. Here too, try to focus on the methods that are relevant to the data. This should describe the methodology employed for the study or research.

Methods, if they are not in the README file:

F.5. Provide a list of Subjects (aka keywords, descriptors). REQUIRED

Please provide a list of descriptors or keywords describing what the data is about, etc. If there are terms from controlled vocabularies or other sources, please indicate (i.e.., gene names, species, chemical names)

Provide a list of subject(s) (or keywords, descriptors) here. To separate keywords, put a comma between each word or phrase.

Whenever possible, use terminology from established thesauri, taxonomies, etc. and make note of where the term is from.

If taxonomies are not available for your research area, simple terms are okay.

Example of a simple list of terms:

Polychlorinated biphenyls, exposure, inhalation, toxicology

F.6. For Iowa Superfund Research Program data, provide the .csv and JSON + XML files for your selected PCB congeners and/or derivatives. REQUIRED for ISRP.

(a) Go to the PCB Identifiers Web Application, at https://apps.iowasuperfund.uiowa.edu/pcb-ids/

- Instructions on how to select theare provided via the “How-To Guide” tab.

(b) email the **Full csv**, and **JSON + XML zip** files to the data curator (brian-westra@uiowa.edu).

(c) Add the **Basic csv** file to your deposit,

- List and describe it in the README
- Use the description provided in the example below, where the **Basic csv** file is renamed to: **03_Substance_Descriptors_and_Identifiers.csv**

README example excerpt:

03_Substance_Descriptors_and_Identifiers.csv

Computed descriptors (IIUPAC Name, InChIKey, SMILES) and database identifiers (CAS Registry Number, DSSTox Substance ID, PubChem CID), if available, and our internal names (ISRP Name) for the substances represented in this dataset.

- ISRP Name: Name of the substance as used by the Iowa Superfund Research Program (ISRP).
- CAS Registry Number: The American Chemical Society's Chemical Abstracts Service (CAS) The Registry Number is a unique identification number, assigned by the Chemical Abstracts Service in the US to every chemical substance described in the open scientific literature, in order to index the substance in the CAS Registry.
- DSSTox Substance ID: An identifier used by the EPA Distributed Structure-Searchable Toxicity (DSSTox) Database.
- InChI Key: InChI Key is based on the InChI (International Chemical Identifier), which is a standardized way of representing the structure of a chemical compound as a text string. Moreover, the InChI Key is a shortened version of the InChI that can be used to identify a chemical compound in a more compact and convenient way.
- IUPAC Name: The International Union of Pure and Applied Chemistry (IUPAC) Name for a chemical substance, which is generated based on rules set by IUPAC.
- PubChem CID: The compound ID (CID) assigned to each substance in the PubChem information repository.
- SMILES: The Simplified Molecular Input Line Entry System (SMILES) is a linear text that is generated by an algorithm to describe the structure of a chemical substance

F.7. Provide citations with DOIs or links for any related works (articles, software/code, datasets) planned for publication or already published. REQUIRED

Provide the full citation (including DOI, ISBN, etc.), so we can add that to the record and link the two together.

For instance, if:

- the dataset is a subset of a published dataset, or incorporates data from other sources,
- code or software is published elsewhere and associated with the dataset
- another article has been published on this dataset
- sequence data is deposited in GEO

Related work (citation, with DOI)	Type of relationship to the data: source of data; publication about the data; subset of the data, etc.)

G. Publishing the Data – Activating the DOI

Contact us when you want the data to become publicly accessible (we activate the DOI, and the record in IRO becomes publicly accessible). **This is not reversible**.

When do we do this? You decide:

A. Publish data **before** the associated manuscript is published

   Some journals may require that the data has been published before they will publish the manuscript.

B. Publish the data **at the same time** that the manuscript is published

   Researchers frequently want to wait until the manuscript is published before making the data publicly available.

   Notify us as soon as the manuscript is published and we will activate the DOI. Send us the DOI for the article (or other related materials) and we will link to it in the IRO record.

C. Publish the data **independently of** an article because you want to share the data:

   NIH encourages data sharing, even if there is no associated publication. By publishing the data, you establish precedent for the data, which is a product of research, and citable.